Fiscal Year:	FY 2018	**Task Last Updated:**	FY 11/13/2018
PI Name:	Koochesfahani, Manoochehr Ph.D.		
Project Title:	ZBOT Thermometry and Velocimetry Diagnostics Development		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48824	**Congressional District:**	8
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	02/05/2016	**End Date:**	12/31/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/31/2019 per NSSC; original end date was 2/14/2018 and had been extended to 7/14/2018 in Feb 2018 and then to 12/31/2018 in June 2018 (Ed., 3/28/19) NOTE: End date changed to 12/31/2018 per NSSC; original end date was 2/14/2018 and had been extended to 7/14/2018 in Feb 2018 (Ed., 7/31/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Pouya, Shahram Ph.D. (Michigan State University)		
Grant/Contract No.:	NNX16AD52A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In-orbit or on-surface cryogenic storage for propellant and life support consumables is essential to all stages of NASA's Exploration Program. However, heat leakage from the surroundings results in liquid vaporization and significant mass loss and/or significant increase in the storage tank pressure. To mitigate the unwanted effects of cryogen vaporization, the Zero Boil-Off (ZBO) concept is being considered as an innovative approach for ventless means of controlling tank pressure through the application of active heat removal and forced liquid jet mixing. The Zero Boil-Off Tank (ZBOT) experiment is designed for operation on the International Space Station (ISS). Its purpose is to examine the ZBO concept and, along with computational modeling efforts, provide information on the fundamental physics of the underlying complex thermofluids environment, which is needed for engineering design optimization of the ZBO concept. The ZBOT experiment currently relies on Digital Particle Image Velocimetry (DPIV) to obtain flow field information and visualization in the test fluid, Perfluoro-n-Pentane (PnP). Difficulties have been encountered, however, in using DPIV. These difficulties include a drastic loss of particles over time and decrease of fluorescence intensity over time from the fluorescent-coated particles, along with concerns over particle sticking and agglomeration. The proposed effort is motivated by the need to obtain non-intrusive, whole-field, temperature and/or velocity data using optical diagnostics in the ZBOT microgravity experiment in order to supplement the computational modeling efforts in progress. Because of the challenges experienced by the ZBOT team in using particle-based techniques (i.e., DPIV), our focus will be on development of molecular-based approaches for measurements within PnP. These include laser induced fluorescence (LIF) methods for thermometry and molecular tagging approaches for velocimetry (MTV).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	(Ed. note October 2018: compiled from PI's technical report submitted May 2018.) This project was launched nearly two years ago with the objective of developing non-intrusive molecular-based temperature and/or velocity measurement methods suitable for the Zero Boil-Off Tank (ZBOT) microgravity experiment. This was partly motivated by the potential challenges connected to the use of particle-based techniques (i.e., Digital Particle Image Velocimetry, DPIV). In addition, whole-field thermometry capability for the ZBOT experiment did not exist at the time. At the conclusion of this two-year effort, we have now demonstrated the feasibility of molecular tagging velocimetry (MTV) and also laser induced fluorescence (LIF) thermometry based on solubilized quantum dots (QD). A major challenge in this project was the identification and/or synthesis of suitable molecular tracers that dissolve in the fluid medium in the ZBOT experiment, Perfluoro-n-Pentane (PnP). The photochemical properties of identified tracers were then characterized to ensure they are possible targets for either MTV, or LIF thermometry. Proof-of-concept experiments needed to be carried out to ascertain feasibility of quantitative measurements, followed by demonstration experiments in the ZBOT breadboard setup which has geometric characteristics similar to the actual ZBOT test section used on the ISS.
Bibliography Type:	Description: (Last Updated: 03/02/2026)
Abstracts for Journals and Proceedings	Pouya S, Blanchard G, Koochesfahani M. "Molecular based optical diagnostics development for thermometry/velocimetry for the ZBOT experiment." 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017